

FC Stack LC 52.30

FC Stack LC 30.30

FC Stack LC 22.30

Liquid-cooled PEM fuel cell stack

–User Manual–



Version 1.5

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1. Introduction

1.1 Foreword

Thank you for buying a product from balticFuelCells GmbH. We wish you every success in the implementation of your product and look forward to working with you in a spirit of trust.

Please read this user manual carefully before using the device. Please observe in particular the information on the intended use and on installation.

Should you nevertheless have any questions concerning installation and operation, please contact the Service team at balticFuelCells GmbH.

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1. Introduction

1.2 Intended use

The FC Stack LC xx.30 is a PEM **fuel cell stack** for providing electrical and thermal energy. The supplied energy carrier, hydrogen, is converted into electricity, heat and water in a catalytic process.

The FC Stack LC xx.30 **may only be used in an overall system** for providing electrical and/or thermal energy. **Hydrogen** (quality: at least 3.0) must be used as the energy carrier.

The device **may only be used as part of an overall system**. The specified technical boundary conditions (see chapter 3.2.) must be observed. The overall system must ensure safe operation.

Within the scope of the technical specifications (see chapter 2.3.), the device can be employed in stationary, quasi-stationary and mobile overall systems.



Improper use of the device can result in **high safety risks, personal injury and property damage and will waive the warranty. A device with visible damage or defects must not be put into operation!**

1. Introduction

1.3 Safety instructions

Read the operating manual carefully and follow all the safety instructions. Keep the operating manual together with the device or as part of the documentation for the overall system



If possible, use only **the recommended tools and original balticFuelCells accessories** for installation. **Immediately disconnect the device from the hydrogen supply if an obvious defect is recognised.** Modifications to the device and incorrect installation or operation jeopardise safety and will result in the loss of the operating permit and voiding of the warranty.



Do not operate the device at ambient temperatures above 50°C. During operation, **avoid direct contact with water, solvents and other chemicals.** For care of the device, please observe our **cleaning and service instructions.**



Please observe the **specifications for the installation space and ambient conditions** (chapter 3.2.). **Keep away from sources of ignition. Do not smoke when handling the device.** During normal operation, the device emits small quantities of hydrogen that can reach an explosive gas concentration in (small) enclosed spaces. The emissions must be able to escape unhindered from the installation space into the environment. **Damage to the device or hydrogen supply can cause hydrogen to escape, resulting in a risk of fire/explosion!**



Store the device, switched off, preferably in a **frost-free room and at temperatures below 50°C.**



The surface of the device becomes hot during operation. **Avoid touching the device.** During operation, the stack has a voltage of up to 52 volts (FC Stack LC 52.30) at the electrical terminals. Ensure that this does not result in any hazards.

In addition to these safety instructions, be sure to also observe all texts in **bold type** to avoid hazards for yourself and others.

1. Introduction

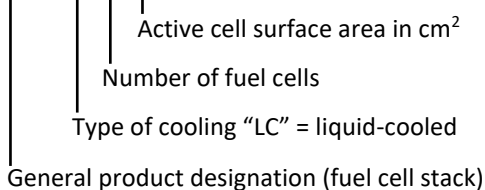
1.4 List of terms and abbreviations

Fuel cell	-	Electro-chemical energy converter that converts energy carriers into electrical energy.
Overall system	-	Operational device for end users. Here this comprises a hydrogen storage facility and, where necessary, devices for conditioning the unregulated power output of the stack.
Hydrogen	-	Energy-rich gas

PEM	-	Proton Exchange Membrane (type of fuel cell)
FC	-	Fuel Cell
LC	-	Liquid-cooled
e.g.	-	for example
incl.	-	including
max.	-	maximum
min.	-	minimum
Fig.	-	Figure
A	-	Ampere
V	-	Volt
W	-	Watt
DC	-	Direct Current
h	-	hours
min	-	minutes
s	-	seconds
g	-	gram
m	-	metre

Explanation of the type code:

FC Stack LC 30.30



1.5 Disposal

Packaging



To ensure that your new device arrives in perfect condition, it is protected by special packaging. All materials used are environmentally friendly and recyclable.

We recommend that you keep the packaging for possible storage and transport of the device. Should you nevertheless wish to dispose of the packaging, please help us to dispose of it in an environmentally friendly manner.



The device

Old devices are not worthless waste. balticFuelCells GmbH will take back the device and thus ensure environmentally friendly disposal of materials and recovery/recycling of valuable raw materials. The device (including all accessories) must therefore **be returned to balticFuelCells GmbH, preferably in the original packaging, at the end of its service life or in the event of a defect.** The shipping costs will be refunded. Please contact balticFuelCells Customer Service for details of the current take-back conditions.

2. Configuration

2.1. Scope of supply

Standard scope of supply:

- FC Stack LC xx.30 1
- Operating manual 2
- Polarization curve

Optional accessories:

- Plastic connecting hose (6x) 3
- Electronic control unit for PEM stacks 4
- Various peripherals

1



2



3

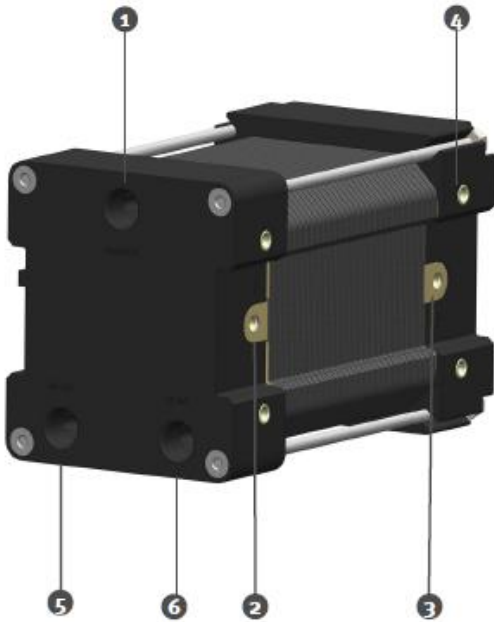


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2. Configuration

2.2. Overview (illustration & description)



View from the front

- 1 - Cooling - inlet (G 1/4")
- 2 - Electrical terminal – anode
(minus symbol) (M5 thread)
- 3 - Electrical terminal – cathode
(plus symbol) (M5 thread)
- 4 - Mounting points (8 x M5 thread)
- 5 - Air outlet (G 1/4")
- 6 - Hydrogen outlet (G 1/4")



View from the rear

- 7 - Hydrogen inlet (G 1/4")
- 8 - Air inlet (G 1/4")
- 9 - Cooling – outlet (G 1/4")

2. Configuration

2.3. Technical specifications

Product	FC Stack LC 22.30	FC Stack LC 30.30	FC Stack LC 52.30	Unit
Number of cells	22	30	52	-
Voltage	11-22	15-30	26-52	VDC
Power Output (nominal)	440	600	1040	W
Hydrogen consumption at nominal power approx.	8.0	11.0	19.0	SLPM
Dimensions (L x W x H)	125 x 85 x 95	147.5 x 85 x 95	208 x 85 x 95	mm ³
Weight (empty) approx.	1.2	1.4	2.0	kg

Amperage

30 A at as basic system

37.5 A at nominal power as extended system

max. 40 A

Hydrogen quality

min. 3.0

Air quality

filtered air without solid, liquid or gaseous impurities

Service life

2000 h with 200 start/stop cycles

Optimum operating temperature

60 - 75°C (stack), depending on the system configuration

Emissions

water (damp air), heat

Installation & storage conditions:

Operating altitude

up to max. 1,500 m above sea level

Ambient temperature

5 – 45°C

Operating temperature

55 – 75°C

Storage temperature

0 – 50°C

Interfaces & electrical connections:

Hydrogen inlet

G 1/4" female thread

Hydrogen outlet

G 1/4" female thread

Hydrogen inlet pressure

250 - 350 mbar

Air inlet

G 1/4" female thread

Air outlet

G 1/4" female thread

Cooling - inlet

G 1/4" female thread

Cooling- outlet

G 1/4" female thread

Power outlet

M5 female thread

3. Installation

3.1. Preparation for installation

Qualification & preparation

Installation of the stack in an overall system should be carried out only by **qualified personnel**. The wearing of **suitable workwear** is recommended.

Caution: Loose clothing and long hair can be caught by moving and rotating parts (e.g. drill). Take care during installation that **no small parts falls into the openings of the stack**. This could result in malfunctions.

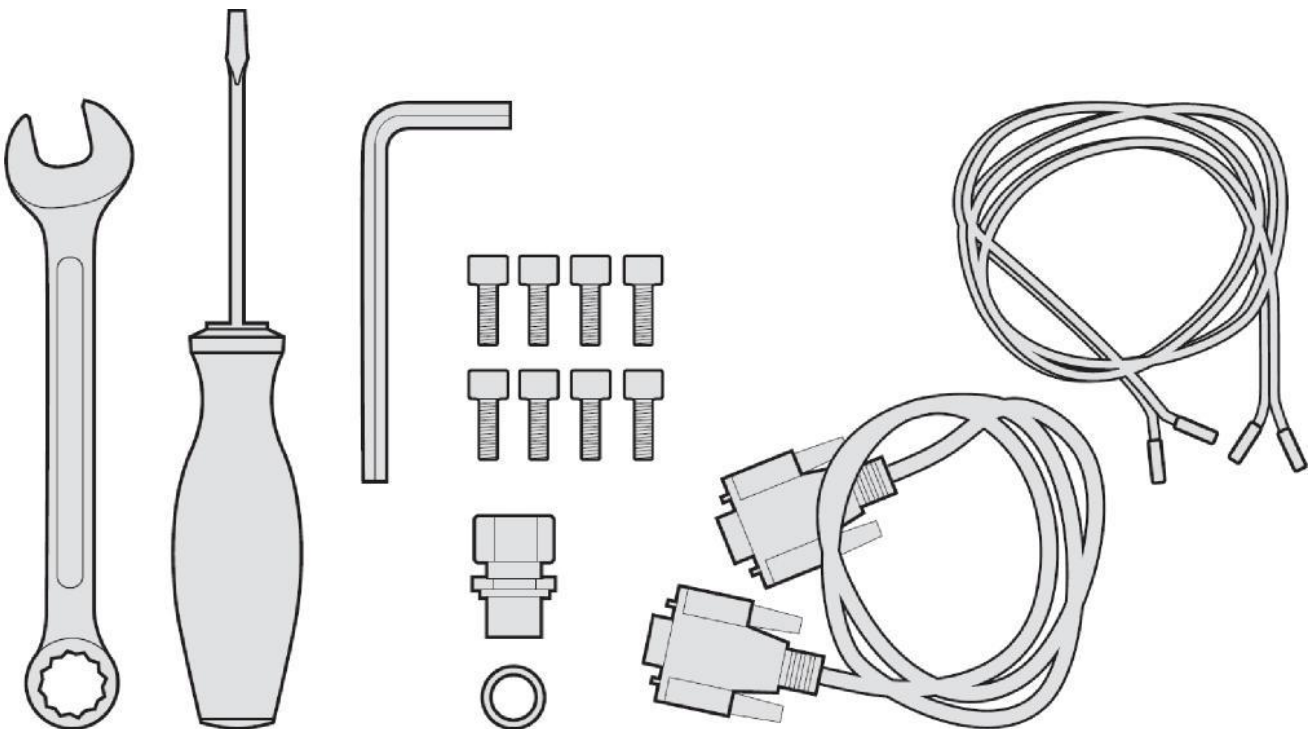
When selecting the installation space, ensure a **permissible ambient temperature range between 5°C and 45°C**.



Tools & accessories

balticFuelCells recommends the following tools and accessories for proper installation of the device:

- Hexagon socket spanners (Allen keys)
- Screwdrivers
- Open-jaw/ring spanners
- 1/4" fittings for media inlets and outlets
- 3-4 hexagon socket screws (Allen) M5
- Suitable power cables (cross-section min. 2.0 mm²) with cable lugs



3. Installation

3.2. Installation space & ambient conditions



The device must be securely mounted in an overall system.

The device may only be operated if there is no explosive mixture of oxygen (including air) and hydrogen in the overall system and its surroundings.



The device must be protected from temperatures above 55°C. During operation, the inlet air flow must not exceed a temperature of 35°C.

1. Minimum dimensions & position

The installation space must not be smaller than the following dimensions (see table):

	FC Stack LC 22.30	FC Stack LC 30.30	FC Stack LC 52.30	
Length	160	200	250	mm
Width	100	100	100	mm
Height	100	100	100	mm



(air out at the bottom)

The device may only be operated in the positions shown. The permissible tilt angles are:

- a) Temporary: 20° to the transverse axis, 20° to the longitudinal axis
- b) Permanent: 10° to the transverse axis, 10° to the longitudinal axis

2. Supply air & discharge of emissions

The device requires fresh air, which must be supplied via the “air in” port.

The device generates waste heat, which must be dissipated to the outside. In closed overall systems, ensure an adequate opening, e.g. for the exhaust air.

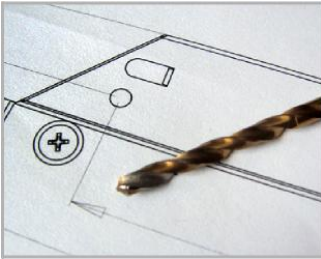
3. Free space for installation of the device and the interfaces

The installation space must allow the individual lines for hydrogen, air, cooling circuit and the electrical connections to be installed and the necessary tools to be used. In the event of a fault, the stack may emit small quantities of hydrogen. Ensure that no ignitable/explosive mixtures of hydrogen and air form in the overall system.

3. Installation

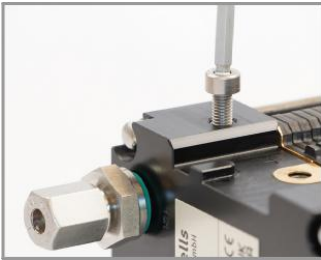
3.3. Installation of the device

The individual installation steps prior to commissioning the device are summarised below:



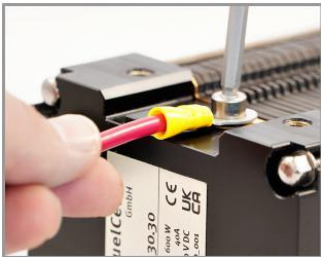
1. Preparation of the installation space

- Drilling of the mounting holes
- Cleaning of the installation space



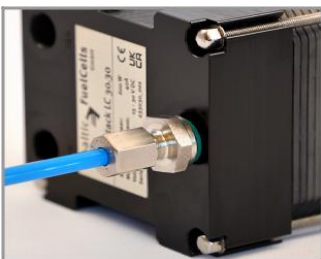
2. Securing of the device

- Fixing the device at up to 4 mounting points (see point 2.2) - Installation of brackets for the media routing, if necessary
- Make sure that no metallic or electrically conductive objects can fall onto the stack



3. Electrical connection

- Connect power outlet terminal using a cable with cable lug (M5 female thread)



4. Media connection

- Preparation of the hose/pipe fitting
- For the water circuit and the hydrogen supply, balticFuelCells recommends a hose/pipe inside diameter of 4 mm, for the air supply a hose/pipe inside diameter of at least 8 mm

3. Installation

Use only screw diameters as specified in this operating manual to install the stack and tighten them only using suitable tools. If you have any questions about the proper use of an individual fastening solution, please contact a balticFuelCells GmbH Service employee.



The installation of the device must ensure permanent and stable positioning within the overall system. Possible **loads (tensile/compressive) on the end plates and terminals must be prevented**. The device should be fixed at not less than three mounting points.

3. Installation

3.4. Electrical connection

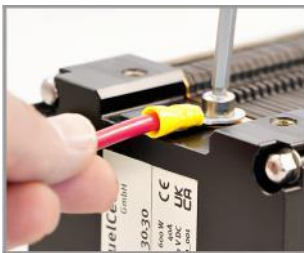


Observe the applicable regulations (e.g. VDE) when working on electrical systems. Incorrectly laid or incorrectly dimensioned cables can result in fires.

All connected cables must have sufficient insulation or dielectric strength and the contact points must be touch-proof. The installation of bare metal cables and contacts is not permitted.

Check the polarity (see point 2.2) before connecting the device.

3.5.1. Power outlet connection



The stack supplies an unregulated output voltage. To discharge the electrical power, M5 female threads are provided for connection of a cable with cable lug. It is essential to observe the information on the downstream electrical components. These should be suitable for the voltages generated by the FC Stack LC XX.30 (please refer to chapter 2.3) and have a continuous load capacity of 38 - 40 amps (input side).

Only cables with a suitable cross-section may be used for power transmission! The cross-section must be calculated on the basis of the expected current.

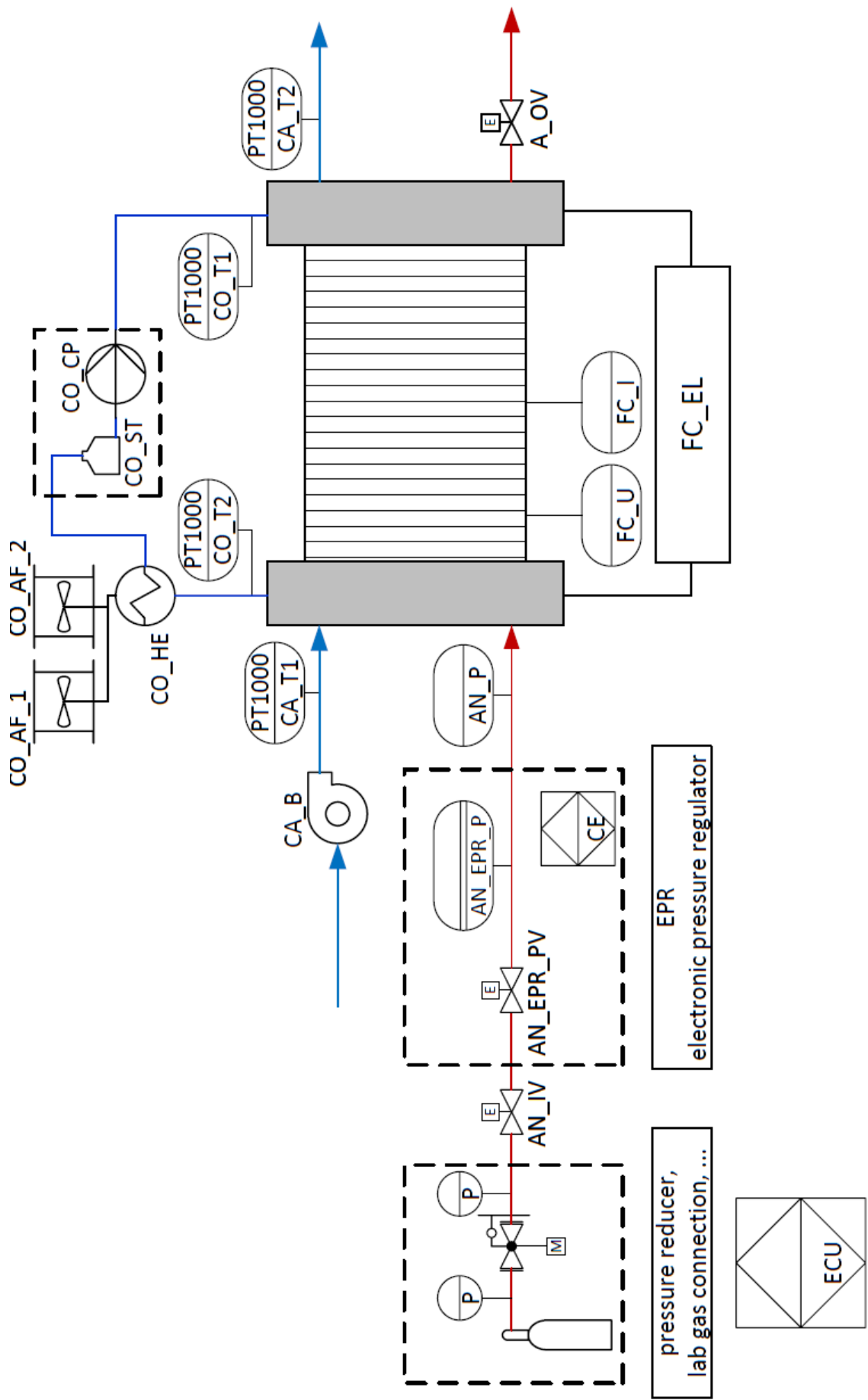
First attach the cable to the negative pole of the stack and connect it to the downstream electrical components. Then proceed in the same way with the positive terminal.



Pay particular attention to the correct polarity of the electrical connections! Avoid short-circuits! These would cause serious damage to the stack and render the device inoperable.

Never disconnect or replace a cable during operation!

3.5. P&I diagram



3. Installation

3.5.1 Description of the P&I diagram

Abbreviation	Designation
ECU	Electronic Control Unit
AN_IV	Hydrogen inlet/safety valve
AN_EPR_PV	Pressure control valve of the EPR
AN_EPR_P	Pressure sensor of the EPR
AN_P	Pressure measurement at hydrogen inlet to FC stack
CA_B	Fan / centrifugal fan
CA_T1	PT1000 temperature sensor – cathode inlet side
CA_T2	PT1000 temperature sensor – cathode outlet side
CO_T1	PT1000 temperature sensor – cooling 1
CO_T2	PT1000 temperature sensor – cooling 2
CO_HE	Coolant heat exchanger
CO_AF_1	Fan 1 heat exchanger
CO_AF_2	Fan 2 heat exchanger
CO_CP	Coolant circulation pump
CO_ST	Coolant expansion tank
A_OV	Hydrogen outlet valve
FC_I	Full cell stack current measurement
FC_U	Full cell stack voltage measurement
FC_EL	Electronic/electric load

3. Installation

3.6. Connection of the media

The FC Stack LC xx.30 has three media inlets and three corresponding media outlets.

Hydrogen supply

The required hydrogen inlet pressure of 250 - 1000 mbar must be ensured. Avoid pressure difference above 300 mbar between anode and cathode side of the stack. balticFuelCells recommends dead-end operation of the FC Stack LC xx.30. In this case, a purge valve should be opened at the outlet every 7.5 seconds. The opening time should be approx. 150 ms. Please note that different valves may also have different opening times. The inlet and outlet valves should have a nominal diameter of at least 1.5 mm. Humidification of the gas is not required.



Connection of a hydrogen supply with a pressure of more than 1000 mbar can lead to serious damage to the stack and to a dangerous leakage of hydrogen. All parts of the hydrogen supply line (fittings, pipes, gaskets, etc.) must be suitable for the maximum permissible pressure. Hydrogen escaping due to leaks can result in dangerous gas accumulations.

Air supply



The cathode of the stack must be supplied with at least a **lambda value of 2**. The calculation must be carried out according to Faraday. A **pressure drop of 20 - 100 mbar** occurs across the stack. This must be considered when choosing the fan.

For the air supply, a hose with an inner diameter of at least 12 mm is recommended for the FC Stack LC 52.30.

Cooling

Cooling must be effected by a circulation pump above a current of 5 amps. During operation, the inlet temperature should be at least 2 Kelvin and no more than 10 Kelvin below the outlet temperature. Overheating of the stack leads to the fuel cell membranes drying out and thus to a reduction in performance. In severe cases, overheating can also cause a malfunction (leakage).

A hose with an inner diameter of 12 mm is recommended for cooling.



Only deionised water (electrically non-conductive) and water-glycol mixtures may be used for cooling of the stack. If the coolant in an overall system is in contact with metals, balticFuelCells recommends the use of **corrosion inhibitors** (e.g. COFERZOL). Please contact us if other coolants are to be used!

4. Operation

4.1. Breaking procedure for FC Stack LC xx.30



Before starting the stack, ensure the following ambient conditions:

- ☒ Ambient temperature between 5°C and 35°C
- ☒ No hydrogen accumulation in the overall system
- ☒ All interfaces connected to the device as described above
- ☒ Hydrogen and air supply assured

Be sure to observe the following instructions when using the stack for the first time or after a long standstill time (more than 3 months). If the instructions are not followed, this will lead to a reduction in performance and a reduction in the service life of the stack.

The full power is not yet available when starting the stack for the first time. The electrodes must initially be started in a defined break in process. Each individual stack has been tested and checked for proper function by balticFuelCells before delivery. The following steps must be performed to bring the stack up to full power:

Step	Description	FC Stack LC 22.30	FC Stack LC 30.30	FC Stack LC 52.30	Unit
1	Ensure supply of air and hydrogen				-
2	Set a load current of 1 ampere » Minimum voltage	15.5	21	36	VDC
3	Wait approx. 30 seconds » The voltage of the stack should have increased				-
4	Set a load current of 2 ampere » Minimum voltage	15	20.5	35.5	VDC
5	Wait approx. 30 seconds >> The voltage of the stack should have increased				-
6	Gradually (1 ampere) increase the current up to 5 ampere; now switch on the circulation pump (CO_CP) » Minimum voltage	14	19	33	VDC
7	Repeat the steps up to 10 ampere » Minimum voltage	12.5	17.5	30.5	VDC
8	Repeat the steps until the voltage is <=	11.5	15.5	26.5	VDC
9	Allow the system to cool down (rest phase; min. 2 hours)				-
10	Repeat the steps, if necessary				-
11	The FC Stack LC xx.30 is now ready for operation.				-

4. Operation

4.2. Normal operation of the FC Stack LC xx.30

Before starting the stack, ensure the following ambient conditions:

- ☒ Ambient temperature between 5°C and 35°C
- ☒ No hydrogen accumulation in the overall system
- ☒ All interfaces connected to the device as described above
- ☒ Hydrogen and air supply assured

Check all other media supplies and first measure the open cell voltage (OCV). Refer to chapter 2.3 for technical specifications. **Do not** increase the current abruptly to the desired operating point, but either select small current increases (max. 1 ampere) or ramp up the current with a maximum increase of 4 amperes per minute. Ensure that the limit values for current and voltage are observed (max. 40 A, FC 22.30: min. 11V; FC 30.30: min. 15V; FC 52.30: min. 26V). If a limit value is exceeded, bring the FC Stack LC xx.30 to a safe condition again. (For example: switch off the load, disconnect the media supply)



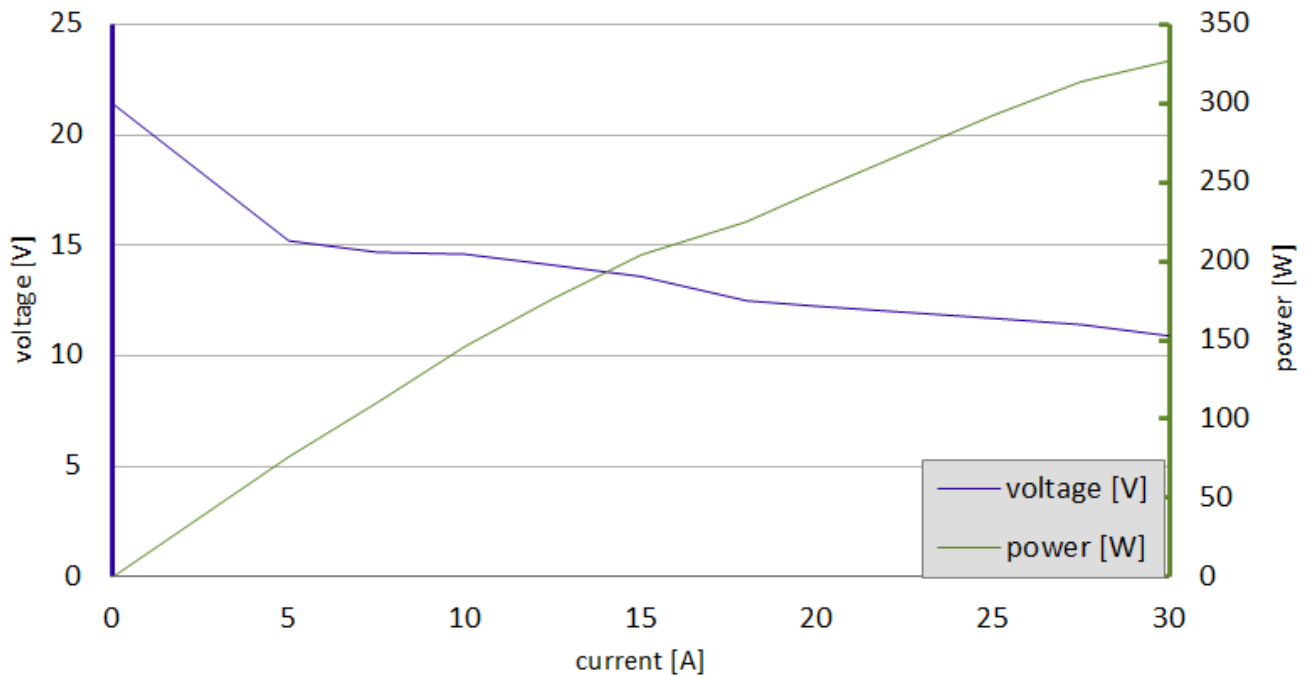
Avoid long operating phases at OCV! This seriously reduces the service life of the stack. balticFuelCells recommends limiting the period of operation at OCV to max. **two minutes per operating cycle**.

To switch off the stack, first disconnect the load from the stack. Then switch off the hydrogen supply and disconnect the stack from the other media.

4. Operation

4.3. Operating characteristics– Basic system

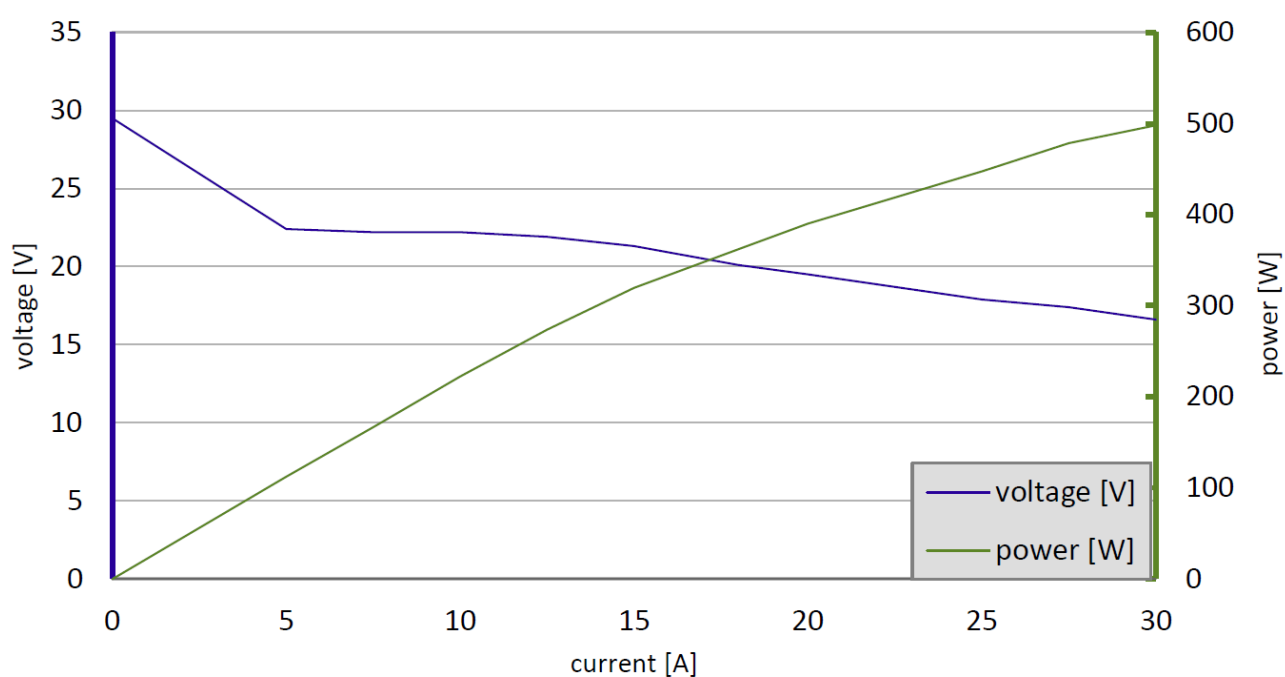
4.3.1 Characteristics FC Stack LC 22.30 – Basic System



Operating conditions:

Hydrogen quality:	3.0
Hydrogen supply:	320 mbar (dead end operation with purging)
Air supply:	Ambient air with fan
Moisture content of H2 and air:	none
Stack operating temperature:	50 - 60°C
Ambient temperature:	22°C
Relative humidity:	44.3%

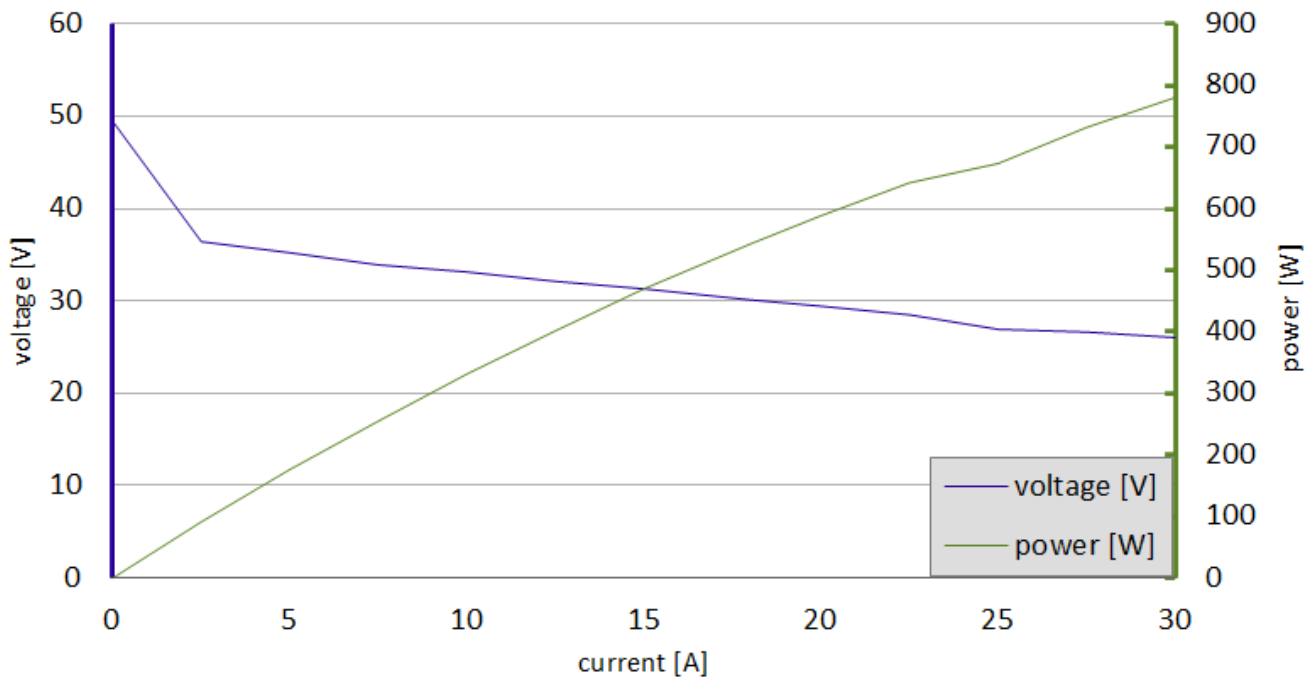
4.3.2 Characteristics FC Stack LC 30.30 – Basic System



Operating conditions:

Hydrogen quality:	3.0
Hydrogen supply:	320 mbar (closed-centre operation with purging)
Air supply:	Ambient air with fan
Moisture content of H2 and air:	none
Stack operating temperature:	50 - 60°C
Ambient temperature:	22°C
Relative humidity:	54.1%

4.3.3 Characteristics FC Stack LC 52.30 – Basic System

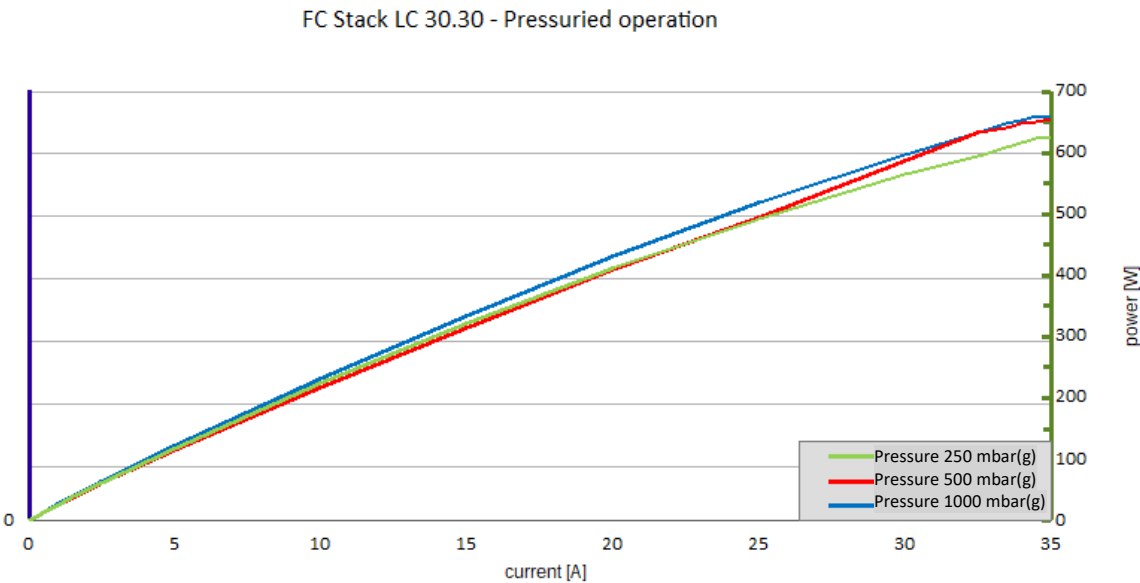


Operating conditions:

Hydrogen quality:	3.0
Hydrogen supply:	320 mbar (closed-centre operation with purging)
Air supply:	Ambient air with fan
Moisture content of H2 and air:	none
Stack operating temperature:	50 - 60°C
Ambient temperature:	20°C
Relative humidity:	47.1%

4. Operation

4.4 Operating characteristics – Extended System



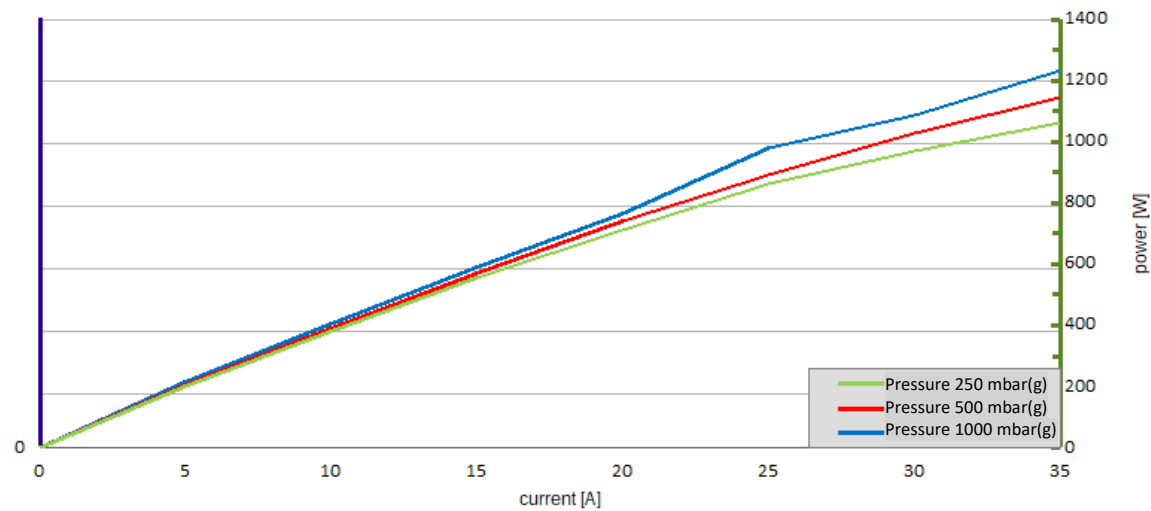
Operating conditions Operating curve – Extended System:

Hydrogen quality:	3.0
Hydrogen supply:	250 - 1000 mbar(g) - Stoichiometry 1.2
Hydrogenrelative humidity :	30%
Air supply:	250 - 1000 mbar(g) - Stoichiometry 2.5
Airrelative humidity :	50%
Stack operating temperature:	65 - 75°C

4. Operation

4.4.1 Characteristics FC Stack LC 52.30 – Extended System

FC Stack LC 52.30 - Pressurized operation



Operating conditions Operating curve – Extended System:

Hydrogen quality:	3.0
Hydrogen supply:	250 - 1000 mbar(g) - Stoichiometry 1.2
Hydrogenrelative humidity :	30%
Air supply:	250 - 1000 mbar(g) - Stoichiometry 2.5
Airrelative humidity :	50%
Stack operating temperature:	65 - 75°C

4. Operation

4.5 Shutdown, storage & transport

If the overall system is not to be used for a prolonged period (longer than 3 months), the FC Stack LC xx.30 should be completely dismantled, sealed with the screw plugs and kept in a suitable storage location.

It is best to store the device in the original cardboard box supplied or in a sufficiently padded, dust and moisture-proof container. Clean the device before packing it. The device should **ideally** be stored at a room temperature of 5 – 25°C.

If the stack is to be stored in the overall system and the temperatures fall below 0°C, ensure that all accumulations of water are drained from the stack. When using water-glycol mixtures for cooling the stack, ensure that the temperatures do not fall below the freezing point of the mixture and that an expansion tank is installed in the overall system.

5.1 Service & after-sales support



Do not open the device! Unauthorised opening endangers safe operation and results in the warranty being waived. The device contains no parts that you can serve or repair yourself.

The Service staff of balticFuelCells GmbH are available to answer your questions, in particular on the following topics:

- Fields of application & ambient conditions of the device
- Installation of the device
- Faults during operation of the device
- Maintenance of the device
- Accessories for the device
- Other abnormalities in the device and the overall system

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Mo – Fr, 8:00 a.m. – 4:00 p.m. (except on public holidays)

To enable us to process your enquiry quickly and efficiently, please provide us with the following information:

- Your customer number
- Serial number of your device
- Description of the fault

5. Maintenance

5.2 Cleaning the device



Switch off the stack and shut off the media supply before cleaning.

Clean the device with a mild cleaning solution applied lightly to a soft cloth.

Use only surfactant-free cleaning agents. Check the air and hydrogen supply.

Do not use compressed air to clean the device!

Also inspect the device for any visible damage while cleaning the device.

6.1 Accessories for hose

Straight push in fitting for G 1/4":

Material: nickel-plated brass Gaskets: NBR

Straight push in fitting G 1/4"-6mm	IQSG 146 G MSV (for 6 mm hose)
Straight push in fitting G 1/4"-8mm	IQSG 148 G MSV (for 8 mm hose)
Straight push in fitting G 1/4"-10mm	IQSG 1410 G MSV (for 10 mm hose)
Straight push in fitting G 1/4"-12mm	IQSG 1412 G MSV (for 12 mm hose)
Or as a version in material POM with the ending FDA (e.g.: IQST 146 G FDA)	

T push-in connectors:

Material 1.4404 Gaskets: FKM

T push-in fitting 6 mm IQST 60 ES	(for 6 mm hose)
T push-in fitting 8 mm IQST 80 ES	(for 8 mm hose)
T push-in fitting 10 mm IQST 100 ES	(for 10 mm hose)
T push-in fitting 10 mm IQST 120 ES	(for 12 mm hose)
Or as a version in material POM with the ending FDA (e.g.: IQST 60 FDA)	

Hose:

Festo plastic tubing 6 mm PUN-H-6X1-BL	(in various colours)
Festo plastic tubing 8mm PUN-H-8X1.25-SW	(in various colours)
Festo plastic tubing 10 mm PUN-H-10X1.5-BL	(in various colours)
Festo plastic tubing 12 mm PUN-H-12X2-BL	(in various colours)
Hose cutter Clip 0 – 14 mm SAS 14	
Silicone hose 12 x16, transparent SILI1216	

Threaded nozzle G1/4":

Threaded nozzle G1/4"-10 mm GT 1410 ES	(for 10 mm hose)
Threaded nozzle G1/4"-12 mm GT 1412	(for 12 mm hose)
Threaded nozzle G1/4"-13 mm GT 1413 ES	(for 13 mm hose)

Gasket made of hard PVC for threads G 1/4" DR 14 K

Wire retention clamp, 16,4 mm, clamping range 15,9 to 16,8 mm DNSS 16,4

Screw-on hose screw connection (coolant pump)

Screw-on hose screw connection G 3/8"-13 (1/2")mm AST3813

These components can be ordered from e.g. www.landefeld.de.

Sealing plugs

GPN 737 G1/4 PA6 yellow with O-ring

These components can be ordered from e.g. www.poeppelmann.com.



The components listed on this page are only recommendations. Other components can also be used, depending on your application.

6. Appendix

6.2 Accessories for pipe

Pipe fittings for G1/4":

Material 316 stainless steel

SS-CM-ML6-NS4-EC79	(for 6 mm pipe)
SS-CM-ML8-NS4-EC79	(for 8 mm pipe)
SS-CM-ML10-NS4-EC79	(for 10 mm pipe)
SS-CM-ML12-NS4-EC79	(for 12 mm pipe)

EC-79 certified for use in hydrogen powered vehicles.

Pipe:

Pipe 6 mm SS-TMP-6MM-1.0	(in various lengths)
Pipe 8 mm SS-TMP-8MM-1.0	(in various lengths)
Pipe 10 mm SS-TMP-10MM-1.0	(in various lengths)
Pipe 12 mm SS-TMP-12MM-1.0	(in various lengths)

These components can be ordered from e.g. www.fitokgroup.com.



The components listed on this page are only recommendations. Other components can also be used, depending on your application.

6. Appendix

6.3 Imprint

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